🔄 Java Loops Summary

Loops are control structures used to execute a block of code repeatedly until a specified condition is met. They are essential for iterating over arrays, processing collections, or performing repetitive tasks.

1. The for Loop (Counter-Controlled)

The for loop is typically used when you know exactly how many times you want to repeat the action, often for iterating a fixed number of times or accessing elements in a fixed-size array by index.

Part	**Description**
Initialization	Executed once at the start; declares and sets up the loop counter (e.g., int i = 0).
Condition	Checked before each iteration; the loop continues as long as this is true (e.g., i < 5).
Iteration	Executed after each iteration; typically increments or decrements the counter (e.g., i++).

Example

```
for (int i = 0; i < 3; i++) {
    System.out.println("Iteration: " + i);
}

// Output:
// Iteration: 0
// Iteration: 1
// Iteration: 2
```

2. The while Loop (Condition-Controlled)

The while loop is used when the number of iterations is unknown, and the loop continues as long as a specific boolean condition remains true. The condition is checked **before** the loop body executes.

Example

```
int count = 0;
while (count < 3) {
    System.out.println("Current count: " + count);
    count++; // Remember to update the condition variable inside the loop!
}

// Output:
// Current count: 0
// Current count: 1
// Current count: 2
```

3. The do-while Loop (Guaranteed First Run)

Similar to the while loop, but the condition is checked **after** the loop body executes. This guarantees that the loop will run at least once, regardless of whether the condition is initially true or false.

Example

```
int runOnce = 10; // This condition (runOnce < 3) is false
do {
    System.out.println("This runs at least once: " + runOnce);
    runOnce++;
} while (runOnce < 3);

// Output:
// This runs at least once: 10
```

4. The for-each Loop (Enhanced Loop)

The enhanced for loop is designed specifically to easily iterate over all elements in an array or a Collection (like ArrayList), without needing to manage the index counter.

Syntax

```
for (DataType element : collectionOrArray) {
    // Code block that uses 'element'
}
```

Example

```
String[] names = {"Leo", "Mia", "Zoe"};

for (String name : names) {
    System.out.println("Hello, " + name);
}

// Output:
// Hello, Leo
// Hello, Mia
// Hello, Zoe
```